

Highlights

Forecast-informed reserve–SOC supervision for drilling-rig power systems: confirmatory simulation and public real-data checks

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- Storage-reserve supervision coordinates batteries and model-reachable generation.
- Frozen 12-seed tests show lower EENS than three matched references.
- Reliability gains require higher realized operating cost.
- Real-load tests improve on persistence at 15 and 60 min in 8/8 series.
- Real BESS and drilling records expose both support and transfer limits.

Forecast-informed reserve–SOC supervision for drilling-rig power systems: confirmatory simulation and public real-data checks

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ABSTRACT

Abrupt drilling-load transitions can exhaust storage reserve before additional diesel generation becomes model-reachable under commitment and ramping constraints. We evaluate a reserve–SOC supervisory model predictive control architecture (RSS) that couples validation-residual load scenarios, conditional value at risk, discrete generator commitment, and storage-reserve supervision. In a prospectively frozen 12-seed simulation study, RSS reduces mean expected energy not served (EENS) by 5.24 kWh relative to rule-based control, 15.59 kWh relative to point-forecast mixed-integer predictive control, and 5.49 kWh relative to risk-adaptive residual-CVaR control. Seed-cluster bootstrap 95% intervals are $[-9.43, -1.29]$, $[-21.45, -9.98]$, and $[-8.70, -2.80]$ kWh, respectively; all three contrasts remain significant after Holm correction. Mean realized operating cost is nevertheless 153.83–280.17 yuan higher, establishing a reliability–cost trade-off rather than joint improvement; no equal-cost sweep was performed. A public-data audit spans eight dataset families and covers load forecasting, BESS behavior, and drilling-process timing. Confirmatory forecasting on eight measured electrical series from three dataset families improves on persistence for 5/8 series at 1 min and 8/8 at both 15 and 60 min. Two real BESS datasets show 95.28–97.98% power–SOC directional consistency. A pooled coarse-state diagnostic for two public drilling records yields shorter dwell times than the simulator (median 2 versus 43 min), although 85.39% of the Energistics labels are unknown. The results therefore support a conditional reliability advantage in the frozen simulation and bounded real-data component checks, but not economic superiority, closed-loop field efficacy, or direct transfer of the simulator's temporal assumptions.

1. Introduction

Drilling rigs are energy-intensive industrial systems whose electrical demand changes sharply with operating state. Drilling, circulation, connection, tripping, and auxiliary activities impose different demand levels and different transition dynamics. When a rig is supplied by a weak grid connection, diesel generator units, and a battery energy storage system (BESS), the operating problem involves more than reducing energy cost. The controller must also maintain power balance, respect generator commitment and ramping constraints, preserve sufficient storage state of charge (SOC), and protect high-priority loads when available supply becomes insufficient.

This problem has a direct operational consequence. A controller that reacts only to instantaneous load may discharge the BESS before a transition, or may start too many diesel units and replace relatively inexpensive grid power with fuel consumption. Conversely, a controller that focuses only on economic dispatch may leave insufficient reserve for a short high-power event. The resulting trade-off is particularly important for drilling rigs because their load profile combines sustained operation with short, high-amplitude transitions.

Energy management studies have investigated hierarchical scheduling, multi-objective microgrid dispatch, energy storage coordination, and model predictive control (MPC). For example, bilevel energy management has been used to coordinate security, reliability, storage, and demand-side management (Jokar, Bahmani-Firouzi and Simab, 2022), while predictive energy management has been applied to hybrid vessel grids (Vafamand, Boudjadar and Khooban, 2020). Stochastic and random-setting formulations have also been used for isolated hybrid microgrids with diesel generation and storage (Vergine, Álvarez-Arroyo, D'Amico, Escaño and Alvarado-Barrios, 2022). These studies establish the value of combining forecasting, uncertainty representation, and constrained optimization. They also

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motivate a system-level evaluation that reports drilling-load transitions, discrete multi-generator commitment, storage protection, and emergency load prioritization under one declared protocol.

The closest drilling rig precedent considered rule-based generator scheduling together with battery storage for diesel power-plant fuel-efficiency improvement (Pavković, Sedić and Guzović, 2016). This prior work motivates the present application, but it also sets a clear comparison requirement: the contribution cannot be reduced to using a BESS in a drilling rig. The relevant question is whether forecast uncertainty and supply risk can be transferred into a controller that remains feasible under the stated active-power constraints while avoiding unnecessary generator commitment.

This paper evaluates a reserve–SOC supervisory energy-management framework for a grid–diesel–battery drilling rig power system. The systems question is whether storage reserve and model-reachable generator capacity can be coordinated early enough, using only causal forecast information, to reduce shortage during abrupt load transitions. The framework links short-horizon forecast scenarios to conditional value-at-risk (CVaR) dispatch, discrete generator commitment, SOC hysteresis, and priority-based load protection. The individual ingredients are established in prior work. The contribution claimed here is a system-specific supervisory coupling whose mechanism and operating effects are tested under a common causal and physical boundary. The paper makes three contributions:

1. We design and implement a reserve–SOC supervisor that combines a causal forecast envelope, current SOC, and model-reachable generator capacity within a receding-horizon grid–diesel–battery dispatch model.
2. We evaluate the controller under a prospectively frozen 12-seed simulation protocol against rule-based and matched MILP references, with validation-only residual scenarios, seed-level inference, component interventions, and constraint audits that separate reliability from operating-cost effects.
3. We add a multi-source public-data checking layer based on real measurements: two drilling-process records, eight electrical-load series, and two BESS operating records. This layer tests forecast transfer, storage behavior, and simulator assumptions without relabeling those data as target-rig closed-loop evidence.

The controller comparison uses 72 h simulated replays. The revised dispatch protocol declares a new seed queue and retains the first 12 upstream-eligible holdouts; previously inspected seeds are development-only. Public real data provide external evidence for selected inputs and physical relationships, but no public source contains synchronized target-rig bus power and supply-side states suitable for a closed-loop replay. The controller-efficacy claim therefore remains simulation-bounded. Section 3 describes the evidence roles and their boundaries.

2. Related work

2.1. Drilling-rig storage and load-shock control

Storage is not new to drilling-rig power systems. Pavković et al. combined rule-based generator scheduling with a battery energy storage system (BESS) to improve diesel-plant operation under drilling loads (Pavković et al., 2016). Chupin et al. reviewed the associated sizing and operating problem (Chupin, Frolov, Korzhavin and Zhdaneev, 2021), while Huang et al. considered offshore scheduling under uncertain renewable generation and platform demand (Huang, Lan, Hong, Wen and Yin, 2019). More recently, Wang et al. proposed operating-state-dependent dual power allocation and converter-level control for an electrified oil rig, and evaluated the system with full-cycle field power and operating records (Wang, Chen, Shi, Wang, Wang, Zhong and Luo, 2025). That field evidence is stronger than the simulation evidence available in the present study. Our narrower question is whether validation-calibrated forecast-error scenarios, model-reachable multi-unit commitment, and SOC reserve supervision improve constraint-feasible active-power dispatch when all alternatives share one causal replay and plant model.

2.2. Closest predictive-dispatch precedents

The closest precedent is the stochastic economic MPC of Chapaloglou et al., which combines data-driven uncertainty scenarios with mixed-integer unit scheduling for an isolated oil-and-gas power system subject to abrupt, irregular loads (Chapaloglou, Varagnolo, Marra and Tedeschi, 2022). Moretti et al. studied a two-layer predictive dispatch algorithm using field data from an off-grid microgrid (Moretti, Polimeni, Meraldi, Raboni, Leva and Manzolini, 2019); Nelson and Johnson demonstrated real-time microgrid MPC for ancillary-service participation (Nelson and Johnson, 2020). These studies establish both the stochastic-MPC precedent and stronger routes to field or experimental validation. The present study differs in its drilling-load replay, grid–diesel–battery topology, 5-s receding-horizon update, explicit generator reachability, and SOC reserve supervisor. Its main methodological control is a matched ablation: persistence, point-forecast, fixed-reserve, residual-CVaR, risk-adaptive, and full supervisory variants

Table 1

Capability comparison with the closest application and predictive-dispatch precedents

Study	Rig / O&G	Abrupt load	Uncertainty	Multi-unit UC	Evidence
Pavković et al. (Pavković et al., 2016)	rig	yes	no	rule-based	simulation
Wang et al. (Wang et al., 2025)	rig	yes	no	not reported	field records
Chapaloglou et al. (Chapaloglou et al., 2022)	O&G	yes	stochastic	yes	simulation
Moretti et al. (Moretti et al., 2019)	no	variation	forecast	not reported	field data
Nelson and Johnson (Nelson and Johnson, 2020)	no	market	forecast	not reported	experiment
Present study	rig simulation	yes	residual scenarios	four units	simulation plus public real-data external checks

execute through the same dispatch implementation and use the same terminal-SOC rule. Because the plants and data differ, we do not claim head-to-head superiority over the cited systems.

2.3. Uncertainty, reserve, and supervisory MPC

Predictive energy management for microgrids is well established (Hu, Shan, Guerrero, Ioinovici, Chan and Rodriguez, 2021; Abbasi and Baleanu, 2023). Relevant formulations include resilience-oriented multi-scenario reserve control (Tobajas, Garcia-Torres, Roncero-Sanchez, Vazquez, Bellatreche and Nieto, 2022), stochastic MILP (Franke, Schneider, Weitzel and Rinderknecht, 2020), and robust cost–risk scheduling with CVaR (Ji, Huang, Xie, Zhou and Zhou, 2018). We use the auxiliary-threshold CVaR formulation of Rockafellar and Uryasev (Rockafellar and Uryasev, 2000), but treat it as a standard ingredient rather than a contribution. The study-specific additions are: (i) full-horizon residual trajectories drawn only from validation data; (ii) a finite-scenario tail-resolution gate that prevents a nominal confidence level from collapsing onto too few scenarios; (iii) risk-conditioned reserve and commitment supervision; and (iv) separate reporting of operating cost, terminal-energy adjustment, and reliability penalty. Priority-based load allocation is retained only as a shortage diagnostic; it is not evidence that total supply adequacy has improved.

2.4. Forecasting role and evidence boundary

The forecasting stage compares naive, tree-based, recurrent, convolutional, and transformer-based candidates. Temporal Fusion Transformer, PatchTST, and iTransformer provide relevant attention-based forecasting precedents (Lim, Arik, Loeff and Pfister, 2021; Nie, Nguyen, Sinthong and Kalagnanam, 2023; Liu, Hu, Zhang, Wu, Wang, Ma and Long, 2024); the implemented candidate set and selection rule are stated explicitly in Sections 4.1 and 5.2. These models are baselines, not the claimed novelty. The controller is evaluated by causal simulation replay, whereas Wang et al., Moretti et al., and Nelson and Johnson provide field or experimental controller evidence (Wang et al., 2025; Moretti et al., 2019; Nelson and Johnson, 2020). We use public real measurements for three bounded external checks: drilling-process timing, cross-domain electrical-load forecasting, and BESS power–SOC behavior. The drilling records lack synchronized rig-bus active power and supply-side state, and the electrical and storage records are not from the target rig. Accordingly, the paper claims simulation-based controller efficacy plus bounded real-data component and assumption checks; it does not claim deployment readiness, voltage/frequency stability, converter-level performance, or target-rig field generalization. Within that boundary, the contribution is the controlled comparison of how causally calibrated forecast uncertainty, discrete commitment, and SOC supervision interact under rapid load transitions.

Table 1 makes the comparison boundary explicit. “Not reported” means that the cited article does not use that element as a reported evaluation feature; it is not a judgment that the element could not be added to the method.

3. System boundary and data

3.1. System boundary

The system contains four functional components: (i) an external grid connection with time-varying available capacity, (ii) a set of diesel generator units, (iii) a BESS with power and SOC limits, and (iv) an electrical load representing the drilling rig and auxiliary equipment. The controller receives measurements and causally available

state information. It produces a receding-horizon plan and applies the first control action before the next measurement update.

The plant is a quasi-static active-power and energy model. It represents source availability, unit commitment, minimum stable output, ramping, start synchronization delay, storage energy and SOC-dependent power limits, but not reactive power, bus voltage, frequency dynamics, short-circuit strength, protection coordination, converter electromagnetic dynamics, or sub-5-s transients. Consequently, “feasible” below always means feasible under these stated model constraints.

The load is partitioned into critical, essential, and interruptible classes for evaluating priority-based protection when available supply is insufficient. The class proportions are control-study settings; deployment values must be approved through the drilling rig’s electrical and safety procedures.

The implementation uses a 5 s sampling interval for the main high-frequency data path. The forecast task uses a 10 min history to predict the following 1 min horizon. The dispatch controller operates on the forecast package and current equipment state; all future true load and future-derived operation-state flags are excluded from online features.

3.2. Evaluation data

Table 2 separates the evidence sources and their roles. Each simulator seed produces a 72 h active-power record at 5 s resolution (51,840 timestamped rows before quality processing). The mechanistic drilling-load and multi-source power-system simulator records total load, operating state, available supply, generator status, battery power, and SOC. One prespecified development seed is used for forecast-model comparison, residual-scenario calibration, controller debugging, and the pre-freeze runtime gate. It is not included in the primary controller inference. The primary dispatch evidence comprises the first 12 upstream-eligible seeds in a queue frozen before any of their controller outcomes were generated.

The development forecasting replay contains 51,709 samples, of which 51,697 are scored. A fixed chronological split is used for model comparison. The power scaler is fitted on the training partition; ensemble weights and complete forecast-residual vectors use the validation partition; and test targets are excluded from model fitting, calibration, and selection. For each prospective seed, the same upstream fitting and calibration procedure is rerun before its dispatch comparison, without changing the frozen hyperparameters or decision rules.

The public-data collection contains eight dataset families and ten prepared one-minute electrical series, totaling 13,571,131 timestamps; 12,351,203 (91.01%) pass the dataset-specific frozen quality rules, and target values are never imputed. Six REFIT houses and the UCI household record provide directly measured aggregate active power. The Tsukuba load target is derived from measured grid import, photovoltaic generation, and battery power rather than a dedicated load meter. M5BAT and Tsukuba provide real BESS power and SOC trajectories. Energistics Well B and Utah FORGE Well 58-32 provide independent drilling-process records; neither contains the synchronized total active power and supply-side state required for controller replay.

OpenCEM was used during pipeline development and its opened period is retained as exploratory rather than recycled as a confirmatory holdout. Petrobras 3W is retained only as a scope audit because its production-well anomaly labels are not power-system risk labels. These exclusions prevent unrelated real data from being used to manufacture a nominal field-performance score.

Figures 1 and 2 summarize the load features and representative operating-state curves used to define the forecasting and dispatch problem.

3.3. Preprocessing and causality

The preprocessing path canonicalizes timestamps, checks the sampling interval, records quality flags, and imputes only the documented short gaps. The forecast input window is constructed from information available at the prediction timestamp. Transition features are based on history, not on the future operation-state label. This distinction is essential because a future-derived transition flag would make the forecast or controller appear stronger while being unavailable during online operation.

Development cases and holdout cases are separated throughout the evaluation. Controller settings and eligibility criteria are fixed before prospective holdout outcomes are inspected, and no parameter is retuned during the 12-holdout primary comparison. Eligibility is determined before dispatch from the public-calibration audit, operation-state codes, transition flags, and supply regime. The resulting holdouts remain draws from the same mechanistic simulator and are not an external-site dataset.

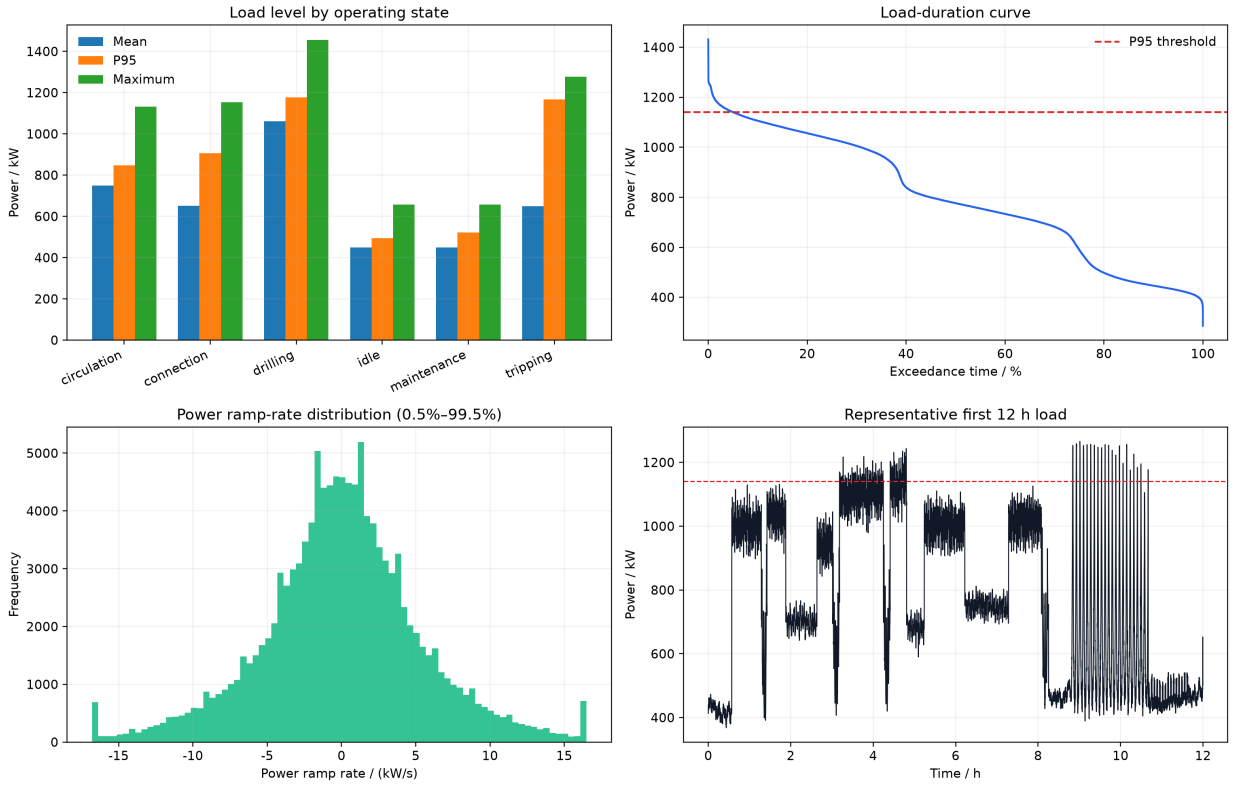


Figure 1: Load-feature summary across operating states and representative intervals

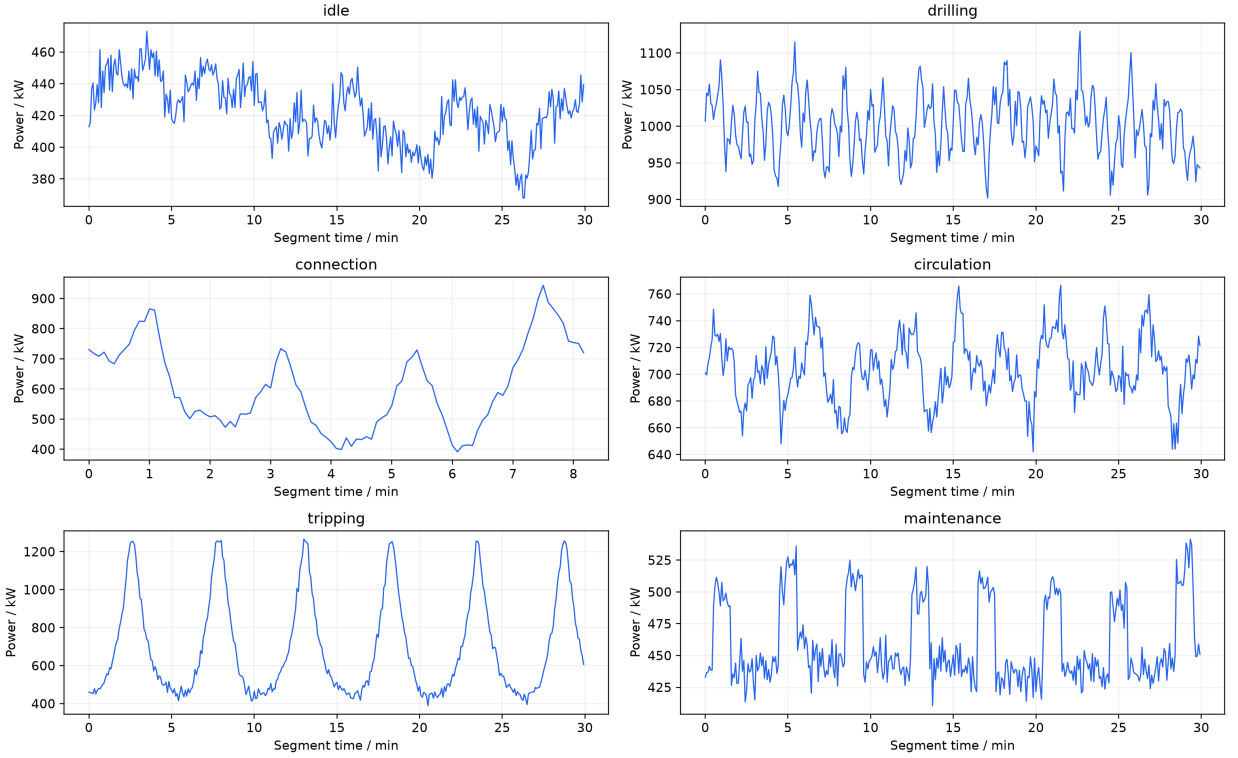
Table 2

Evaluation datasets and their roles

Dataset / source	Role	Resolution	Content and provenance
Development replay	Forecast selection, scenario calibration, and debugging	5 s	One 72 h simulator seed; excluded from primary controller inference
Prospective simulator holdouts	Confirmatory dispatch comparison	5 s	First 12 upstream-eligible 72 h seeds retained in frozen queue order
Energistics Well B; Utah FORGE 58-32	Real drilling-process assumption audit	1 min derived	18,597 and 56,682 state-labeled minutes; no synchronized rig-bus active power
REFIT houses 1–5 and 11; UCI household; Tsukuba	Confirmatory cross-domain load forecasting	1 min	Eight series from three dataset families; direct aggregate power except the measurement-derived Tsukuba load target
M5BAT Pb1; Tsukuba	Observational BESS behavior audit	1 min	Real power, SOC, and available control/setpoint channels; not generated by RSS
OpenCEM; Petrobras 3W	Development/scope audit only	source-dependent	Opened OpenCEM pilot excluded from confirmation; 3W anomaly labels not treated as supply-risk labels

For the public electrical series, raw timestamps and units are normalized by source-specific adapters, duplicate timestamps are rejected, and one-minute records retain an explicit quality flag. The confirmatory forecast protocol uses chronological 60/20/20 wall-clock partitions separated by one full UTC day. Ridge regression and histogram gradient boosting are selected by validation MAE and then refitted on training plus validation data. Persistence and a 60-min rolling mean are fixed baselines. Mean absolute scaled error (MASE), using adjacent quality-passing training minutes as the scale, is the primary cross-dataset metric. The 1-min selected-model-versus-persistence family is adjusted by Holm’s method across eight series; 15- and 60-min results are secondary and retain unadjusted day-cluster bootstrap

Representative load segments by operating state

**Figure 2:** Representative load curves for six drilling operating states

intervals. The test target for every listed series is opened once, and all dataset–horizon results are retained regardless of direction.

The BESS analysis is observational. It tests AC/DC agreement, charge and discharge efficiency, setpoint tracking where available, and whether the sign of adjacent SOC change agrees with the power convention during active periods. Eligible pairs are quality-passing adjacent one-minute records with $|\Delta\text{SOC}| \geq 0.01$ percentage points and active-power magnitude of at least 20 kW for M5BAT or 5 kW for Tsukuba. Directional consistency requires $\text{sign}(\Delta\text{SOC}) = -\text{sign}(P)$ under the adopted discharge-positive convention. An initial M5BAT run produced no eligible rows because a released DC-power field is numerically in kW despite its W suffix. That technically invalid run is retained. The revised normalization was frozen only after annual raw-data checks against current times voltage, and the corrected run is reported as R1. The drilling-process analysis maps source-specific channels to coarse states, aggregates the labels to one minute, and compares pooled contiguous state-run durations with the simulator; it does not infer electrical power from mechanical or hydraulic proxies. Unknown labels are retained as a state rather than removed or bridged. In Energistics, 85.39% of one-minute labels are unknown, so its pooled dwell statistic is an assumption diagnostic rather than a like-for-like state-conditional calibration target.

4. Reserve–SOC supervisory coordination framework

4.1. Forecasting module

Let L_t denote the total active load at time t , and let H be the forecast horizon. The forecasting module produces the point forecast in Eq. (1):

$$\hat{\mathbf{L}}_{t+1:t+H} = f_{\theta}(\mathbf{X}_{t-W+1:t}), \quad (1)$$

where W is the historical input length and $\mathbf{X}$ contains the available load, equipment, and causal process features. The comparison includes Persistence, XGBoost, long short-term memory (LSTM), temporal convolutional network (TCN), PatchTST, iTransformer, and state-aware ensemble variants. The causal error-feedback ensemble supplies the point forecast used by the controller under the frozen protocol. This path was fixed before the prospective controller outcomes were generated; it was not selected by inspecting their test errors. PatchTST and iTransformer provide established transformer comparators (Nie et al., 2023; Liu et al., 2024).

For control, complete 12-step residual vectors are formed from the chronological validation partition only. At each origin, 50 residual vectors are sampled by strata of maximum positive forecast error and added to the causal point forecast:

$$\mathbf{L}_{t+1:t+H}^{(s)} = \max\{0, \hat{\mathbf{L}}_{t+1:t+H} + \epsilon_{1:H}^{(s)}\}, \quad (2)$$

where each residual block $\epsilon_{1:H}^{(s)}$ is sampled as one unit, preserving its within-horizon temporal dependence, and each scenario has probability $1/50$. A chronological first-half/second-half audit within the validation partition assesses interval and upper-tail coverage. Test targets are neither read nor used to fit, rank, rescale, or select these scenarios.

4.2. Supply, storage, and load variables

At each control step, the decision variables include grid power P_t^g , generator power $P_{i,t}^{gen}$, unit commitment $u_{i,t}$, storage charging power P_t^{ch} , storage discharging power P_t^{dis} , and nonnegative unserved-load variables U_t . The BESS SOC is S_t , and η_{ch} and η_{dis} denote charging and discharging efficiencies. For a priority class k , $U_{k,t}$ is the load in that class that is not served, so $0 \leq U_{k,t} \leq L_{k,t}$ and $U_t = \sum_k U_{k,t}$.

The physical balance for a scenario is represented by Eq. (3):

$$L_t^{(s)} - U_t^{(s)} = P_t^{g,(s)} + \sum_i P_{i,t}^{gen,(s)} + P_t^{dis,(s)} - P_t^{ch,(s)}, \quad (3)$$

where $U_t^{(s)}$ is nonnegative unserved load. This convention makes the served load equal to $L_t^{(s)} - U_t^{(s)}$ and is also used when computing unserved energy. The scenario-dependent storage state is updated by Eq. (4):

$$S_{t+1}^{(s)} = S_t^{(s)} + \frac{\eta_{ch} P_t^{ch,(s)} \Delta t_h}{E_{BESS}} - \frac{P_t^{dis,(s)} \Delta t_h}{\eta_{dis} E_{BESS}}, \quad (4)$$

where $\Delta t_h = 5/3600$ h. All scenarios begin from the same measured SOC. The model also enforces SOC bounds, charge/discharge exclusivity, and unit-commitment constraints. The dispatch configuration uses a 5 s control interval, a 30 min evaluation window with a 10 min warm-up, a 1300 kW grid limit, four 240 kW generator units, a 500 kW/133 kWh BESS, initial SOC 0.65, SOC bounds [0.15, 0.90], and charge/discharge efficiencies of 0.95. The unit minimum stable output is 120 kW, the unit ramp limit is 30 kW per control step, and the minimum up/down times are 12/6 steps. A start command reaches synchronized minimum-stable output after six steps (30 s); the execution layer prevents earlier generation. The V24 confirmatory analysis uses this fixed 30-s delay and does not include a startup-delay sweep. Available BESS power is linearly derated within 5 percentage points of the lower and upper SOC limits. The cost configuration uses 0.71 yuan/kWh for grid energy, 7.50 yuan/L for diesel, 0.15 yuan/kWh for storage degradation, 100 yuan/kWh for unserved energy, and 5 yuan per start command. These values define a configured operating-cost proxy rather than a site-calibrated life-cycle cost and should be replaced with site measurements during deployment. The 500 kW/133 kWh storage setting has a nominal power-to-energy ratio of 3.76 h^{-1} ; it is a stress-test configuration, not a value identified from a target-rig asset. The optimizer uses the same public CAT XQP300 50-Hz piecewise fuel curve used to report fuel consumption: 33.8, 47.3, and 62.5 L/h at 50%, 75%, and 100% load, respectively.¹ The MILP includes the minimum-load fuel intercept and a second segment above 75% unit load. Dispatch is solved with HiGHS 1.12.0 through SciPy 1.17.1, using a 5 s time limit and a 1.00×10^{-4} relative MIP gap for every MILP path. The recorded decision latency covers forecast-to-dispatch computation.

¹Caterpillar, *XQP300 Mobile Generator Sets*, product specification, accessed 15 September 2026: https://www.cat.com/en_ID/products/new/power-systems/electric-power/mobile-generator-sets/1000024412.html.

For each generator unit, the MILP enforces the minimum-stable and rated-power bounds, the ramp limits, and the logical relation between commitment, startup, and shutdown variables. Minimum up/down constraints include the executed transition history at the beginning of each horizon. The first control action is shared across forecast scenarios, whereas later continuous recourse variables may differ by scenario. This prevents the controller from using future scenario identity to choose the first physical action.

4.3. Dispatch objective

The receding-horizon controller minimizes expected operating cost plus a CVaR penalty on shortage cost over the current horizon, as defined in Eq. (5):

$$\min J = \mathbb{E}_s \left[J_{energy}^{(s)} + J_{fuel}^{(s)} + J_{start}^{(s)} + J_{deg}^{(s)} + J_{short}^{(s)} + J_{spill}^{(s)} \right] + \beta \text{CVaR}_\alpha \left(J_{short}^{(s)} \right). \quad (5)$$

Here $J_{short}^{(s)}$ is the shortage penalty for scenario s , $J_{spill}^{(s)}$ penalizes unused surplus, and the scenario probability is $1/S$. The CVaR term is implemented with an auxiliary threshold η and excess variables ξ_s in Eq. (6):

$$\text{CVaR}_\alpha(J_{short}) = \eta + \frac{1}{(1-\alpha)S} \sum_{s=1}^S \xi_s, \quad \xi_s \geq J_{short}^{(s)} - \eta, \quad \xi_s \geq 0. \quad (6)$$

This auxiliary-variable representation is the standard linearizable CVaR construction (Rockafellar and Uryasev, 2000); the stochastic-MILP and robust cost–risk literature provides related scheduling precedents (Franke et al., 2020; Ji et al., 2018).

The scenario set contains 50 equally weighted validation-residual trajectories. The primary CVaR setting is $(\alpha, \beta) = (0.90, 1.00)$; for risk levels 0, 1, 2, and 3, respectively, α is 0.80, 0.85, 0.90, and 0.90, while β is 1.00, 1.00, 1.50, and 2.00. Thus, the smallest represented tail contains five equally weighted scenarios. A pre-run discretization gate rejects any setting with fewer than five effective tail scenarios. The more expensive $S = 100$, $\alpha = 0.95$ setting is reserved for offline sensitivity because development solves approached or exceeded the 5 s control interval. The commitment variables are integer and include generator availability, minimum stable output, ramping, minimum up/down time, and startup/shutdown logic. Every MILP comparison uses the same realized replay, plant, prices, initial state and pending start history, risk-conditioned terminal-SOC rule, horizon, solver backend, time limit, and MIP gap. Predicted horizon CVaR is reported separately from realized accounting. Realized operating cost contains grid energy, diesel fuel, storage throughput, and start commands; terminal-energy adjustment and VOLL times EENS are separate columns. Their sum is called total social cost, not risk-adjusted operating cost.

4.4. Reserve–SOC supervisory layer

The supervisory layer selects the operating mode around the rolling dispatch controller while retaining the constraints of the physical model. It uses current SOC, available capacity, the causal risk signal, and the forecasted supply–demand envelope. In normal operation, the controller uses risk-adaptive CVaR MPC; under protection, it uses a deterministic reserve-envelope MPC (implementation label ML–Robust–MPC). This path does not solve a min–max or uncertainty-set robust-optimization problem: it adds a fixed 10 kW reserve and a disagreement envelope equal to 0.25 times the absolute point-forecast–persistence difference. Protection begins when the risk level is at least 2 or SOC is at most 0.30. Normal operation resumes when the risk level is at most 1, SOC is at least 0.35, and both conditions persist for 12 control steps.

When protection is active and SOC is at most 0.30, the controller requests at least two model-reachable generator units if the forecast-deficit guard is triggered. A grid-headroom rule targets SOC 0.75 and limits the storage adjustment to 20% of rated storage power. This rule preserves reserve without starting an additional generator solely to charge the battery.

The causal four-level risk target is formed from the realized future peak only for offline labeling. A safety margin of $\max(50 \text{ kW}, 0.05 L_{peak})$ is subtracted from available supply, and the resulting margin ratio defines normal, watch, warning, and severe classes at thresholds 0.15, 0.05, and 0. The online classifier receives only current and historical features; future load and future-derived state indicators are excluded. These thresholds and the supervisor thresholds above were frozen before V24 and were not subjected to a confirmatory threshold sweep.

Table 3
Symbols and constraints

Symbol / group	Meaning	Unit	Parameter provenance / configuration
L_t	total rig active load	kW	mechanistic evaluation record
p_t^g	grid active power	kW	1300 kW grid limit
$p_{i,t}^{gen}$	generator i active power	kW	four 240 kW units
$u_{i,t}$	generator commitment	binary	MILP formulation
p_t^{ch}, p_t^{dis}	BESS charging and discharging power	kW	500 kW BESS limit
S_t	BESS state of charge	% or p.u.	initial 0.65; bounds [0.15, 0.90]
U_t	nonnegative unserved load; sum of class-level curtailed load	kW	priority-based allocation setting
$\Delta t_h, H, W$	time step, forecast horizon, history length	h/steps	5/3600 h; 12-step horizon; 120-step history
β, α	CVaR weight and confidence level	—	$\beta = 1.00$ and $\alpha = 0.90$ at base risk; risk-level settings stated in Section 4.3

4.5. Auxiliary risk interface

Earlier development also tested evidence-fusion variants of the four-level risk signal. They changed recognition and probability diagnostics but did not improve energy outcomes under the matched replay. Those negative results and their full definitions are retained in the supplementary reproducibility record; evidence fusion is not part of the V24 method or primary claim.

4.6. Emergency load protection

When total available supply cannot satisfy all loads, the emergency module allocates supply by priority. Critical loads are protected first, essential loads are protected second, and interruptible loads absorb the remaining deficit. The controller reports both critical load unserved energy and total unserved energy. A reduction in critical load shortage does not imply that total shortage has been eliminated.

Table 3 summarizes the principal symbols and configured constraints.

5. Experimental protocol

5.1. Research questions

The experiments address four questions:

RQ1. How do the forecasting candidates compare on total-load, peak, and transition errors under the same replay?

RQ2. Does reserve–SOC supervision reduce EENS, and how much additional realized operating cost does it require relative to rule-based control and matched point-forecast and residual-CVaR MILPs?

RQ3. Which control components change the physical dispatch, and does the controller remain feasible under generator, grid, storage, missing-data, and telemetry disturbances?

RQ4. Which parts of the evidence chain transfer to public real measurements, and which simulator assumptions are challenged by external checking?

RQ3 is examined through leave-one-component-out interventions for residual scenarios, adaptive CVaR, SOC supervision, and generator-first execution, together with retained disturbance diagnostics and model-constraint audits. The confirmatory V24 evidence does not contain a startup-delay, economic-weight, or supervisor-threshold sweep. RQ4 is examined through frozen multi-horizon load forecasting, observational BESS power–SOC checks, and state-dwell comparison against two independent drilling-process records. None of the RQ4 analyses is treated as a causal closed-loop test of RSS.

5.2. Baselines

The forecasting baselines are Persistence, XGBoost, LSTM, TCN, PatchTST, iTransformer, and state-aware or ensemble variants. This set includes a naive reference, classical machine learning, recurrent neural networks, convolutional sequence modeling, and recent transformer-based models. For compact tables, the longer forecasting labels are abbreviated as follows: state-aware patch transformer (SA-PT), causal error-feedback ensemble (CEF), validation-weighted ensemble (VWE), state-aware dual-branch patch transformer (SA-DB), and state-aware TCN-attention (SA-TA).

The primary dispatch paths are rule-based control (RB), persistence MILP, matched point-forecast MILP, fixed-reserve MILP, residual-CVaR MILP, risk-adaptive residual-CVaR MILP, and the full risk–SOC supervisory MILP (RSS). All MILP names describe one implementation path whose inputs or supervisory switches are changed explicitly. RB is retained as a transparent operational reference. Auxiliary evidence-fusion and perfect-foresight studies from earlier development are not pooled with this primary comparison. The three prespecified confirmatory references span distinct questions: RB is the transparent operational comparator, point-forecast MILP is the matched deterministic optimization comparator, and risk-adaptive residual-CVaR is the immediate parent controller without the reserve–SOC supervisory layer. The remaining paths characterize the implementation progression and component mechanisms rather than enlarging the primary hypothesis family.

All online MILPs use the same plant model, realized load and availability, initial SOC and unit history, price terms, terminal-SOC policy, solver, and constraint-audit path. Persistence versus point forecast changes only the forecast source; fixed reserve adds only the declared reserve; residual-CVaR adds validation-calibrated scenarios and CVaR; risk adaptation changes only the risk-conditioned CVaR coefficients; and RSS adds the supervisory state machine and generator-first reserve action. Additional leave-one-out paths remove one of these components from RSS.

5.3. Metrics

Forecast metrics include mean absolute error (MAE), root mean squared error (RMSE), the coefficient of determination (R^2), normalized MAE, sMAPE, peak MAE, and transition MAE. Transition MAE is reported because average error can hide the short events that determine generator commitment and storage reserve.

Dispatch metrics include total and normalized EENS, loss-of-load duration, realized operating cost, terminal-energy adjustment, reliability penalty, total social cost, predicted horizon CVaR, diesel fuel, start commands and synchronizations, minimum SOC, maximum decision latency, and constraint-audit failures. Risk-interface metrics include Macro-F1, balanced accuracy, high- and severe-risk recall, multiclass Brier score, mean ignorance, mean conflict, changed risk levels, and changed physical actions. Constraint audits check active-power balance, dynamic grid/generator/storage capacity, SOC bounds, simultaneous charge/discharge, integer commitment, startup synchronization delay, and solver/runtime validity.

EENS is the sole primary endpoint. Realized operating cost is the principal economic secondary endpoint and excludes both EENS monetization and predicted CVaR. Terminal-energy adjustment and VOLL times EENS are reported separately, so a reliability improvement cannot be restated as an independent cost benefit created by the same penalty.

5.4. Reproducibility protocol

Development and holdout cases are separated before controller comparison. The primary analysis uses the first 12 eligible seeds from the prospectively declared queue and treats the seed as the independent unit, with three 30-min scored operating scenarios clustered within each seed after separate 10-min warm-ups. Thus each seed contributes 1.5 scored hours. Exact two-sided sign-flip tests, 20,000-resample seed bootstrap intervals, and Holm correction across the three primary controller contrasts are applied at seed level. Upstream eligibility is evaluated before controller execution from the public-calibration audit, operation-state codes, transition flags, and supply regime. Seed processing follows the declared queue order; no outcome-based replacement is permitted.

Previously inspected seeds are development-only. The prospective queue is processed in order; eligibility is written before dispatch and is based only on operation-state and supply-regime coverage. Every eligible result is retained regardless of direction, including null, adverse, failed, and timed-out cases. Versioned configurations, ordered seed lists, hashes, and trajectory-level audit results are retained in the working evidence ledger and are required in the updated reproducibility package.

Earlier aborted protocol artifacts remain read-only and are not pooled with the confirmatory analysis. V24 uses a wholly unseen queue (20261120 onward), a full-seed serial-execution lock, explicit failure records, and a 5-s wall-clock

Table 4
Experimental comparison protocol

Item	Forecast comparison	Dispatch comparison
Evaluation basis	Simulated development replay plus eight frozen public real electrical series	Power-system simulation
Sampling	5 s for development; 1 min for public real data	5 s control/evaluation path
Forecast window	10 min history → 1 min in development; causal lags and 1/15/60-min horizons in public data	same forecast package per dispatch comparison
Baselines	11 development candidates; persistence and 60-min rolling mean for public confirmation	RB plus six matched MILP paths; four leave-one-component-out interventions are exploratory
Main split	fixed development boundaries; public series use chronological 60/20/20 wall-clock partitions with one-day gaps	first 12 eligible prospective holdouts; three scenarios clustered per seed
Primary metrics	development MAE; public-data MASE with MAE/RMSE companions	EENS
Additional checks	peak/transition error, day-cluster intervals, runtime	operating cost, loss-of-load duration, SOC, fixed-delay synchronization, latency, and constraint audit
Reproducibility	fixed splits, test-open-once rule, data/config hashes, and software environment	fixed controller settings, seed queue, hashes, seed-level statistics, and constraint audits

gate. If no learned risk candidate passes the prespecified validation gates, an always-severe dispatch-facing fallback is applied without excluding the seed; fallback use is reported and is not described as learned-risk certification.

Table 4 distinguishes the forecasting and dispatch protocols.

The forecast models use 120 historical steps (10 min) to predict 12 steps (1 min), batch size 512, a maximum of 20 training epochs, early-stopping patience of 5, learning rate 8.00×10^{-4} , weight decay 1.00×10^{-4} , and seed 20260902. The replay was run with Python 3.11.16, PyTorch 2.9.1, XGBoost 3.2.0, and an NVIDIA GeForce RTX 5060 Laptop GPU.

6. Results

6.1. Forecasting comparison

Table 5 summarizes the prespecified development replay. The validation-weighted ensemble (VWE) obtains the lowest MAE (19.63 kW), while the causal error-feedback ensemble (CEF) used by the controller obtains 20.25 kW, 63.10% below Persistence. VWE also has the lowest RMSE (41.81 kW) and highest R^2 (0.972); CEF has the lowest peak MAE (21.53 kW). These are development-replay comparisons, not independent field validation and not a basis for changing the frozen prospective controller.

The forecasting comparison serves two purposes. It compares modern sequence models and quantifies the validation residuals used by the dispatch controller, and it quantifies peak and transition errors that affect generator commitment and storage reserve. CEF supplies the causal point forecast, while complete CEF validation-residual vectors supply the uncertainty scenarios. The strongest forecast is an existing iTransformer baseline; the contribution evaluated below is the propagation of calibrated forecast uncertainty into the energy-management decision.

Figure 3 separates overall behavior from a principal target transition and shows four representative predictors: VWE (lowest MAE), CEF (the controller's causal point forecast), PatchTST (a transformer comparator), and Persistence (the naive reference). Figure 4 reports all candidates using the abbreviations defined in Section 5.2.

6.2. Validation-residual calibration and pre-freeze gate

The residual library contains 54,290 complete 12-step vectors from the development validation partition. Residual mean and standard deviation are -0.04 and 43.37 kW, respectively, and adjacent lead-time residuals have correlation 0.842 . In a chronological split internal to validation, the empirical central 80% and 90% coverages are 0.811 and 0.904 ; exceedance of the fitted upper 90th and 95th percentiles is 0.100 and 0.051 . Thus the calibration audit is close to nominal

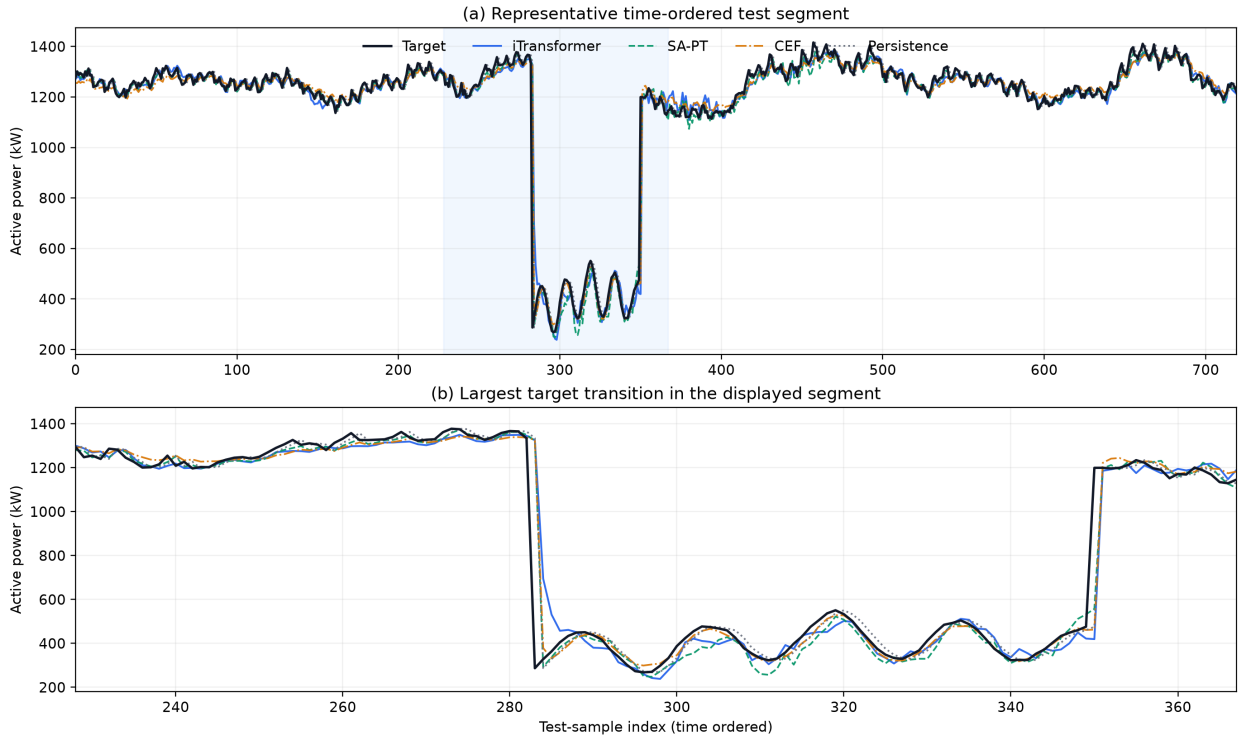


Figure 3: Representative forecast traces and transition detail

Table 5

Forecasting comparison on the evaluation replay

Model	MAE (kW)	RMSE (kW)	R^2	Peak MAE (kW)	Transition MAE (kW)
VWE	19.63	41.81	0.972	22.37	156.17
CEF	20.25	42.81	0.971	21.53	158.70
SA-TA	21.16	43.00	0.970	23.20	156.38
SA-DB	21.27	43.51	0.970	23.92	158.17
SA-PT	21.64	44.14	0.969	23.21	158.15
PatchTST	22.84	47.11	0.964	23.55	165.39
iTransformer	22.84	47.68	0.964	24.35	164.01
TCN	22.93	45.05	0.967	26.64	159.50
LSTM	27.62	49.63	0.961	32.66	163.86
XGBoost	31.54	59.96	0.942	49.95	164.57
Persistence	54.87	97.47	0.848	62.22	195.76

at both central and upper-tail levels without reading test targets. With 50 equiprobable scenarios and $\alpha = 0.90$, the represented CVaR tail contains five scenarios and passes the prespecified minimum-tail gate.

Before freezing, all 21 development method–scenario trajectories passed the active-power, capacity, SOC, charge/discharge, commitment, and 30-s startup- synchronization audits. The largest recorded optimization latency was 3.70 s, below the 5-s control interval. These development checks qualify the implementation for prospective evaluation; they are not included in the primary effect estimates.

6.3. Twelve-seed confirmatory dispatch comparison

All first 12 eligible seeds in the frozen V24 queue completed without a technical failure. Each seed contributes three clustered operating scenarios, and the seed remains the independent analysis unit. Each seed-level value sums three 30-min scored windows (1.5 scored hours) after their warm-ups, and Table 6 reports means of those seed-level values

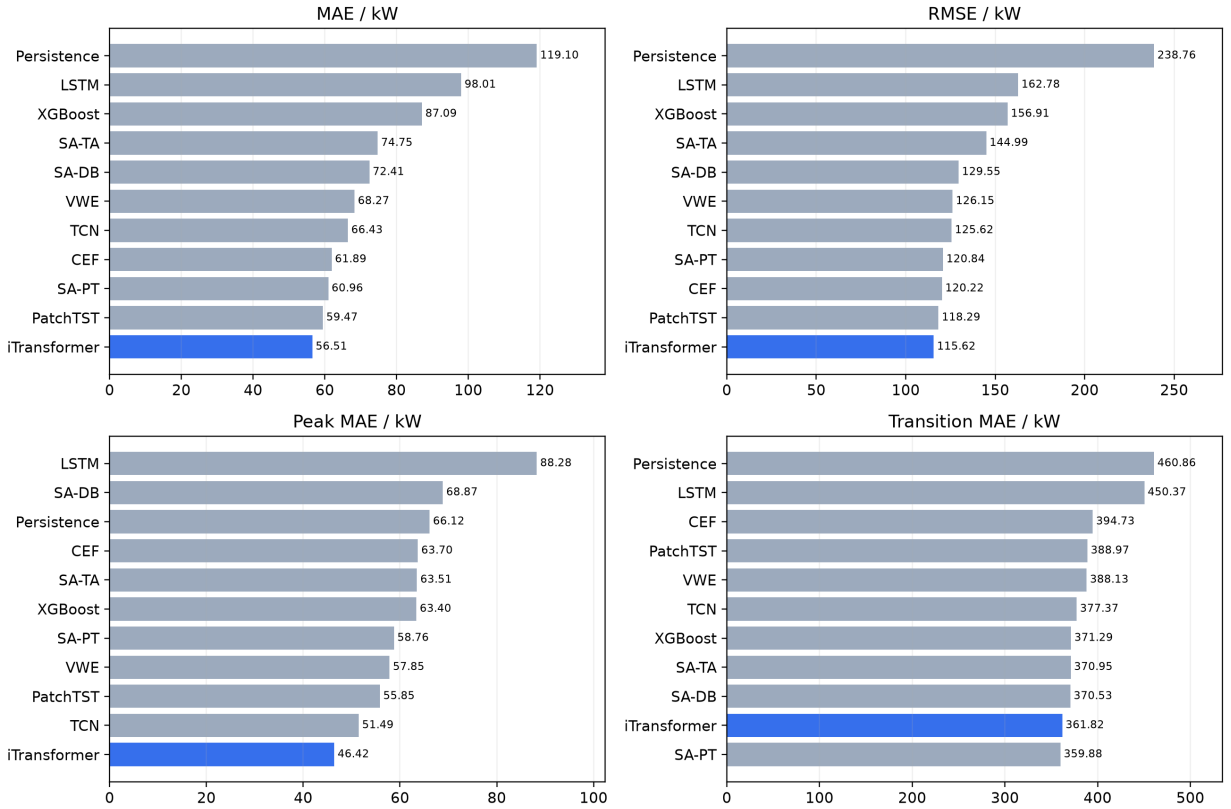


Figure 4: Forecast metric comparison for all candidates

Table 6

V24 confirmatory dispatch results across 12 frozen simulation seeds. Values are seed-level means except the worst decision time. Start-command rates are normalized by 1.5 scored hours per seed.

Method	EENS (kWh)	Operating cost (yuan)	Diesel (L)	Start cmd. (h ⁻¹)	Worst time (s)
RSS	56.35	1,907.15	94.13	68.61	4.80
RB	61.59	1,711.17	71.09	58.50	< 0.01
Risk-adaptive residual-CVaR	61.85	1,753.33	77.34	63.17	4.79
Residual-CVaR	62.97	1,810.77	76.76	71.44	4.60
Fixed-reserve MILP	70.73	1,625.46	64.03	57.44	0.15
Point-forecast MILP	71.94	1,626.98	62.52	58.94	0.14
Persistence MILP	76.66	1,642.28	62.64	61.44	0.16

for all seven matched controllers. RSS has mean EENS 56.35 kWh, compared with 61.59 kWh for RB, 71.94 kWh for the point-forecast MILP, and 61.85 kWh for the risk-adaptive residual-CVaR MILP. These differences correspond to mean EENS reductions of 8.51%, 21.67%, and 8.88%, respectively.

The reliability gains are not free. RSS mean realized operating cost is 1,907.15 yuan per seed, compared with 1,711.17 yuan for RB, 1,626.98 yuan for the point-forecast MILP, and 1,753.33 yuan for the risk-adaptive residual-CVaR MILP. Thus the corresponding mean cost increases are 195.98, 280.17, and 153.83 yuan. RSS also uses more diesel and issues more start commands than these references. The primary result is therefore a reliability–cost trade-off, not simultaneous improvement of both endpoints. Because no matched economic-weight sweep or equal-cost constraint was run, these data do not establish Pareto dominance or an EENS advantage at equal operating cost.

Table 7 reports the three prespecified primary contrasts. All bootstrap intervals exclude zero and all exact sign-flip tests remain significant after Holm correction. The benefit is heterogeneous rather than universal at seed level: RSS

Table 7

Prespecified V24 primary contrasts. Differences are RSS minus the reference; negative EENS favors RSS and positive cost means RSS is more expensive. Intervals are seed-cluster bootstrap 95% intervals.

Reference	Δ EENS (kWh)	95% interval (kWh)	Holm p	Δ cost (yuan)
RB	−5.24	[−9.43, −1.29]	0.0396	+195.98
Point-forecast MILP	−15.59	[−21.45, −9.98]	0.00293	+280.17
Risk-adaptive residual-CVaR	−5.49	[−8.70, −2.80]	0.00586	+153.83

improves 8/12 seeds relative to RB, 11/12 relative to the point-forecast MILP, and 10/12 relative to the risk-adaptive residual-CVaR MILP. Leave-one-seed-out mean differences remain negative for all three contrasts.

Across the 12,960 RSS decisions, wall-clock latency has median 0.009 s, 90th, 95th, and 99th percentiles of 0.828, 1.560, and 2.413 s, and maximum 4.802 s; no recorded decision exceeds the configured 5-s interval. These are development-platform measurements, not an industrial real-time certificate.

The post-freeze audit reports zero configuration-hash mismatches, zero seed-manifest mismatches, and zero technical failures. One seed uses the declared always-severe risk fallback because no learned risk candidate passed its validation gate; that seed remains in every effect estimate and is not treated as a certified learned-risk result. In a post hoc descriptive exclusion of that seed (20261131), mean RSS-minus-reference EENS remains negative for RB (−6.25 kWh; 8/11 seeds favorable), point-forecast MILP (−14.97 kWh; 10/11), and risk-adaptive residual-CVaR (−5.48 kWh; 9/11). This check does not replace the prespecified 12-seed inference. A historical file-matching error affected only the normalized EENS denominator stored for the residual-CVaR method. Raw EENS, the three primary contrasts, and their exact tests are unaffected; the corrected normalized summary is retained in the post-freeze audit.

6.4. Component interventions

The V24 component analysis uses the same 12 seeds and removes one component at a time from the full controller. The reserve–SOC supervisor contrast is one of the confirmatory primary comparisons; the remaining component contrasts were declared before their runs but are interpreted as exploratory. Removing generator-first execution increases EENS by 1.33 kWh per seed on average (full minus removed: 95% interval, −2.37 to −0.44 kWh; exact $p = 0.00586$). Replacing residual scenarios with a point forecast increases EENS by 2.45 kWh (interval, −4.28 to −0.79 kWh; $p = 0.0156$). In contrast, the isolated risk-adaptive CVaR component has a mean difference of only −0.004 kWh, with an interval spanning zero (−0.028 to 0.021 kWh; $p = 1.00$).

These interventions support generator-first reserve execution and residual scenarios as contributors to the simulated reliability effect. They do not support a marginal reliability claim for adaptive CVaR, and they do not identify a single rule as the sole mechanism because the full controller still contains interacting commitment, reserve, and SOC constraints.

6.5. Public real-data component and assumption checks

All 24 frozen electrical dataset–horizon combinations were completed and retained. Table 8 summarizes selected-model performance relative to persistence. At the 1-min primary horizon, 5/8 effects favor the selected model, but only three are Holm-significant improvements; two series show significant degradation and the remaining three are inconclusive after multiplicity correction. The median and equal-dataset-weight mean MASE differences are −0.008 and −0.004, respectively. This is not evidence of a universal 1-min improvement. At 15 and 60 min, all eight series favor the selected model and every day-cluster 95% interval for the MAE difference lies below zero. These secondary results indicate more consistent cross-domain transfer at the longer horizons. The selected model also has lower test MASE than the prespecified 60-min rolling-mean comparator in 8/8 series at each horizon; this is a descriptive secondary comparison, whereas multiplicity-controlled inference is restricted to selected model versus persistence at 1 min. The eight series are not eight independent sources: six are REFIT houses, with UCI and Tsukuba providing the other two dataset families.

Table 9 exposes the dataset-level 1-min results that underlie the aggregate row. HGB denotes histogram gradient boosting.

The two observational BESS audits give complementary physical checks (Table 10). For M5BAT, active same-sign AC/DC power has correlation 0.9991, energy-weighted one-way efficiencies are 0.9867 during discharge and 0.9480 during charge, and 97.98% of eligible adjacent active minutes have a power direction consistent with the SOC

Table 8

Confirmatory forecasting on eight public real electrical series. MASE differences are selected model minus persistence; negative values favor the selected model. Multiplicity correction applies only to the 1-min family.

Horizon	Series	Better	Median Δ MASE	Mean Δ MASE	Interpretation
1 min	8	5	−0.008	−0.004	mixed; 3 significant improvements
15 min	8	8	−0.785	−0.767	secondary; all intervals below zero
60 min	8	8	−1.378	−1.441	secondary; all intervals below zero

Table 9

Dataset-level confirmatory results at the primary 1-min horizon. Differences are selected model minus persistence. Intervals are day-cluster bootstrap 95% intervals for Δ MAE; p_H is Holm adjusted.

Series	Model	Test days	Δ MASE	Δ MAE [95% CI] (kW)	p_H	Result
REFIT 1	HGB	107	−0.0383	−0.0033 [−0.0061, −0.0019]	0.0080	improve
REFIT 2	HGB	111	+0.0246	+0.0018 [−0.0007, +0.0043]	0.3558	inconclusive
REFIT 3	HGB	110	+0.0302	+0.0021 [+0.0012, +0.0028]	0.0080	degrade
REFIT 4	HGB	100	−0.0374	−0.0015 [−0.0021, −0.0002]	0.0720	inconclusive
REFIT 5	Ridge	108	+0.1007	+0.0075 [+0.0070, +0.0086]	0.0080	degrade
REFIT 11	HGB	62	−0.0141	−0.0007 [−0.0013, −0.0003]	0.0080	improve
UCI household	HGB	281	−0.0015	−0.0001 [−0.0005, +0.0002]	0.5357	inconclusive
Tsukuba	Ridge	241	−0.0983	−1.0661 [−1.0894, −1.0397]	0.0080	improve

Table 10

Observational checks on public real BESS records. AC/DC correlation and power– Δ SOC correlation are distinct quantities; signs follow the adopted source convention.

Dataset	Rows	Quality	Eligible min	SOC direction	AC/DC corr.	Power– Δ SOC corr.
M5BAT Pb1	4,254,560	96.55%	219,365	97.98%	0.9991	−0.6568
Tsukuba	1,742,400	99.86%	390,151	95.28%	—	−0.8223

change convention. In Tsukuba, 95.28% of eligible adjacent minutes are direction-consistent and the power–SOC-change correlation is −0.8223. These observations support the physical plausibility of the storage variables used by the framework, but the trajectories were generated by their native site controls rather than RSS.

The drilling-process comparison produces an adverse assumption check. Energetics Well B contributes 18,597 one-minute state records and Utah FORGE 58-32 contributes 56,682. Both have a median contiguous state dwell of 2 min and 95th percentiles of 33.35 and 34.30 min, respectively. The simulator has a median of 43 min and a 95th percentile of 155.65 min, or 21.5 and 4.60 times the corresponding pooled real-data medians. Energetics has 85.39% unknown coarse-state labels, which are retained as contiguous runs; source state mixes also differ. Thus the pooled diagnostic does not support the simulator’s coarse-state persistence distribution, but it is not a like-for-like state-conditional calibration or evidence about measured rig power or controller performance. It limits transportability and motivates real-data-informed temporal recalibration followed by a new controller run.

6.6. Emergency priority-load ablation

The emergency allocation diagnostic separates protection priority from total supply adequacy. It compares proportional curtailment with a priority-greedy allocator under the same stress scenarios. The two policies produce the same total unserved energy in every scenario, while priority protection transfers shortage away from critical loads toward essential and interruptible loads. In the base case, critical-load unserved energy decreases from 219.45 to 11.29 kWh; under the joint extreme stress, it decreases from 1,034.90 to 231.16 kWh. The corresponding total unserved energy remains 366.22 and 1,679.26 kWh, respectively.

Load prioritization changes the distribution of shortage, not the amount of available supply. The class fractions used here are control-study settings that require site-specific electrical and safety approval before deployment. Table 11 provides the scenario-level values.

Figure 5 shows the critical-load and total-shortage comparison across the five stress scenarios.

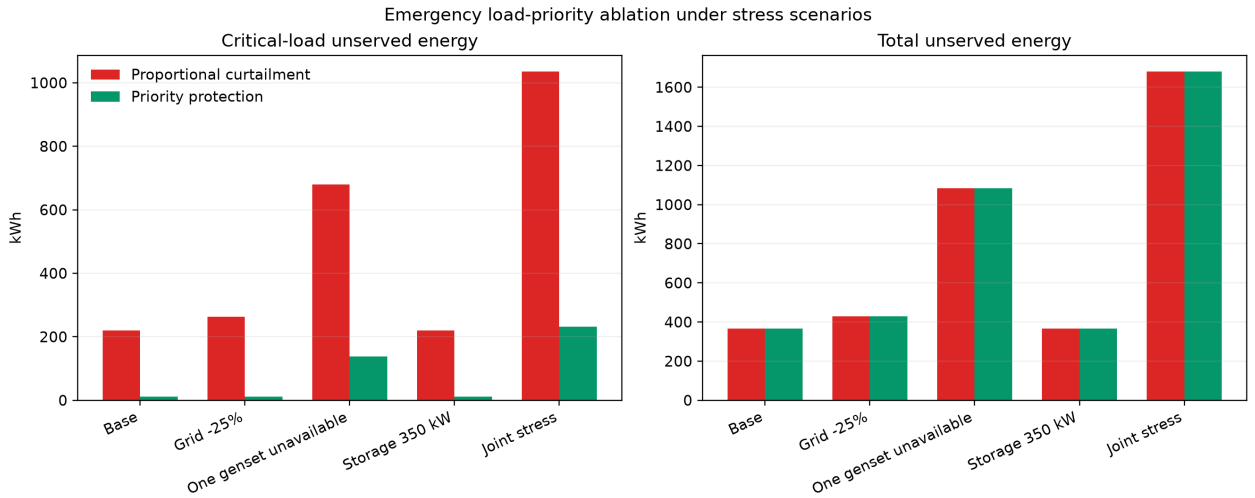


Figure 5: Emergency load-priority ablation

Table 11

Emergency load priority ablation under stress scenarios

Scenario	Critical unserved: proportional (kWh)	Critical unserved: priority (kWh)	Reduction (kWh)	Total unserved (kWh)
Base	219.45	11.29	208.16	366.22
Grid derating, 25%	263.57	11.29	252.28	430.10
One generator unavailable	679.87	137.84	542.03	1,083.29
Storage power cap, 350 kW	219.45	11.29	208.16	366.22
Joint extreme stress	1,034.90	231.16	803.75	1,679.26

6.7. Missing-data and combined-stress diagnostics

The missing-data diagnostic increases MAE from 83.94 kW without injected missingness to 84.33 kW under the moderate condition and 90.66 kW under the severe condition.

Under a combined disturbance involving 25% grid derating, one unavailable generator, a 350 kW storage power cap, a 10 percentage-point SOC bias, a 2.50% load increase, and a 15 s telemetry delay, priority protection reduces critical-load unserved energy from 1,034.90 to 231.16 kWh. Total unserved energy remains 1,679.26 kWh because the allocation layer cannot increase available supply.

Figure 6 shows the detailed joint-extreme trajectory, complementing the cross-scenario comparison in Figure 5.

7. Discussion

7.1. Energy-system performance

Across the 12 frozen simulator seeds, RSS has lower mean EENS than all six reported alternatives. The three prespecified primary contrasts remain significant after Holm correction, and their bootstrap intervals exclude zero. However, four seeds are adverse relative to RB, one is adverse relative to the point-forecast MILP, and two are adverse relative to the risk-adaptive residual-CVaR MILP. The mean operating-cost differences are positive in every primary contrast. The defensible conclusion is therefore a conditional reliability advantage with higher operating cost, not uniform seed-level or economic superiority. In particular, the absence of a matched cost-weight sweep prevents an equal-cost or Pareto-superiority claim.

The public real-data results examine different links in the evidence chain. Forecast transfer is consistent at 15 and 60 min across eight electrical series drawn from three dataset families, while the 1-min result is mixed and includes

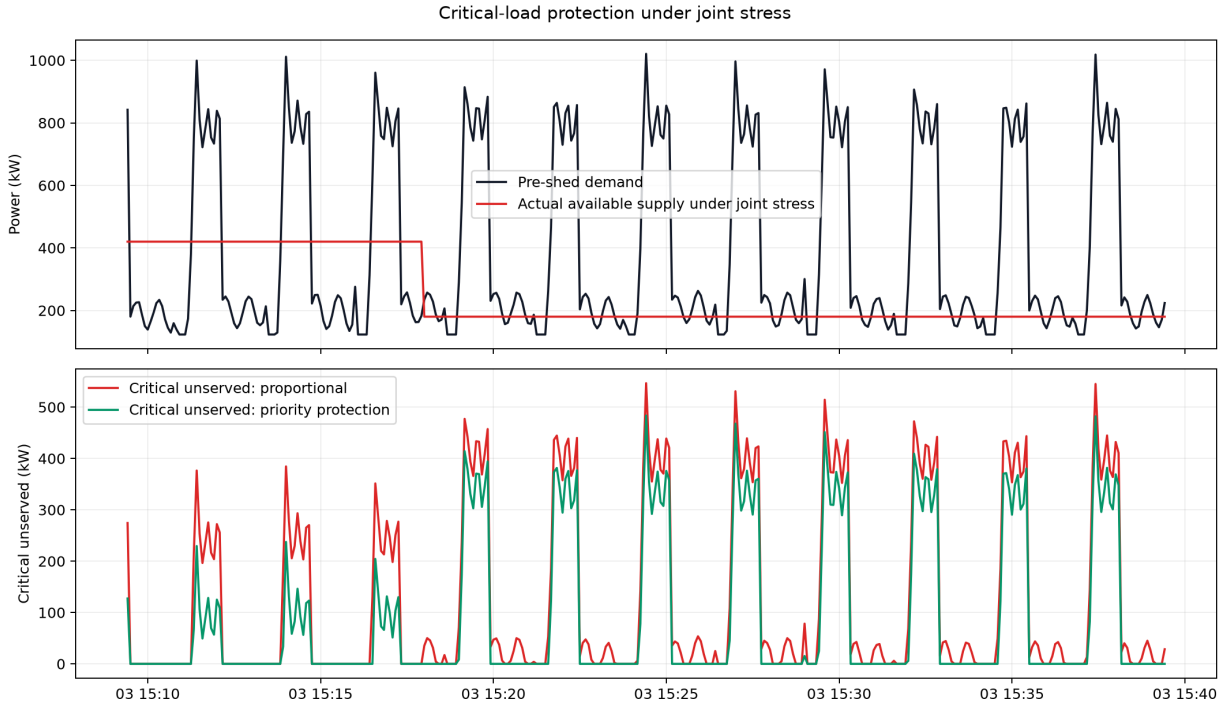


Figure 6: Critical-load protection under joint stress

two significant degradations. Two BESS records support the sign and coherence of power–SOC relationships used by the controller model under the stated activity thresholds. Conversely, the two pooled drilling-process diagnostics challenge the simulator’s state-persistence distribution, although Energistics label coverage limits the comparison. This combination of supportive and adverse findings is more informative than treating every public dataset as another controller benchmark: the data identify which assumptions transfer and which require recalibration.

7.2. Operating mechanisms

The V24 interventions align partially with the proposed mechanism. Removing generator-first execution or replacing residual scenarios with a point forecast increases EENS on average, whereas the isolated risk-adaptive CVaR effect is indistinguishable from zero. The confirmatory full-controller contrast against risk-adaptive residual-CVaR supports the added reserve–SOC supervisory package, but the layer contains interacting rules and should not be presented as a single-rule causal estimate. The fixed supervisor thresholds were not swept, and no multifactor interaction experiment separates their joint effects. Priority-based allocation remains a protection mechanism: it redistributes shortage away from critical loads under severe stress but cannot reduce total shortage when aggregate supply is fixed.

7.3. Engineering applicability

Earlier drilling rig studies established the value of rule-based generator scheduling and battery storage. The present framework adds forecast uncertainty, discrete multi-generator constraints, explicit storage reserve, and priority-based protection in a single rolling controller. The resulting structure is a candidate for evaluation in other industrial grid–diesel–battery systems with abrupt loads and limited grid capacity. The public real-data analyses show that the forecasting and storage-variable interfaces can be exercised on measured records, but they do not establish transfer of the closed-loop policy.

Deployment requires synchronized total active power, individual generator states and outputs, battery SOC and power, grid capacity measurements, fuel consumption, energy prices, and approved load classes. Equipment ramp limits, minimum up/down times, efficiency curves, startup costs, degradation costs, and terminal SOC requirements

should be replaced with site values. The next validation step is an external replay with site data followed by hardware-in-the-loop testing on the intended control platform.

8. Limitations and reproducibility

The controller-performance record is generated by one mechanistic simulator family; synchronized target-rig electrical measurements were not available. The prospective seeds test stochastic robustness within that simulator, not transfer across rigs or sites. They do not capture installation-specific operator actions, protection behavior, or the full communication stack. A separate heterogeneous-simulation protocol was not complete because one of 12 prespecified units had no eligible stable-drilling window before controller execution; the remaining 11 units are not used to manufacture a replacement efficacy conclusion.

The public electrical tests are retrospective cross-domain datasets, not drilling-rig power. Six series share the REFIT dataset family; UCI is a second residential source, and the Tsukuba target is reconstructed from microgrid power channels. The BESS audits are observational and do not compare the native controllers with RSS. Their reported directional consistency depends on the frozen activity and Δ SOC thresholds; no threshold or temporal-aggregation sensitivity was run. OpenCEM was excluded from confirmation because its intended test period had already been opened during development. Most importantly, the real drilling-process records show shorter pooled coarse-state runs than the simulator, but 85.39% of Energistics labels are unknown and the state mappings and mixes are not identical. The result is therefore an adverse diagnostic rather than a direct state-conditional validation. Until the temporal model is recalibrated and the controller is rerun, or the mismatch is bounded by an explicit robustness analysis, the simulation may overstate persistence available to the predictive controller.

The model is quasi-static and active-power only. Its fixed synchronization delay does not reproduce diesel-engine speed, voltage build-up, breaker closing, frequency excursions, converter controls, battery thermal limits, or detailed aging. SOC-dependent BESS power is represented by a simple linear end-region derating, and fuel use by a public steady-state piecewise curve. Therefore, passing the constraint audit is not evidence of electrical stability or safety. The emergency load classes and all equipment and economic parameters require site-specific approval and calibration.

The cost model is a configured proxy containing grid energy, diesel fuel, linear storage throughput, and a 5-yuan start-command term. It excludes demand charges, nonlinear battery aging, maintenance, and site-calibrated hot/cold start wear. The reported start quantity is a model-level command count over three disjoint stress windows, not a field cold-start rate. The 12-seed sample also limits inferential resolution and remains conditional on one simulator family. No confirmatory startup-delay, cost-weight, supervisor-threshold, or BESS-size sweep is available, so the parameter choices and higher switching rate should not be interpreted as economically optimized.

The research implementation treats a missing or over-time optimization result as a technical-gate failure; it does not silently substitute a successful trajectory. No hold-last-command or rule-based switchover was evaluated. Deployment would additionally require a certified, constraint-safe fallback policy and timing tests on the intended controller hardware; the observed development-platform latency distribution establishes neither industrial real-time certification nor fail-safe behavior.

The source artifacts, configurations, result manifests, and analysis scripts are fixed under the public archival tag `v24-v28-reproducibility-20260922`; its code and evidence submodule is pinned to `commit 3499ed3`. Frozen configuration hashes, failed-run records, and compact summaries of the 24 real-data forecast outputs are retained there. The earlier public v1.0.0 package remains a V21 archive and is incomplete for the reported V24/V28 results. Raw third-party datasets remain subject to their original licenses.

9. Conclusions

This paper evaluated whether causal forecast information can be used to coordinate storage reserve with model-reachable diesel generation during abrupt drilling-load transitions. The reserve–SOC supervisor combines forecast-error scenarios with CVaR dispatch, integer generator commitment, and SOC hysteresis within a grid–diesel–battery power-system model.

Across 12 frozen simulation seeds, RSS reduces mean EENS by 5.24 kWh relative to rule-based control, 15.59 kWh relative to point-forecast MILP, and 5.49 kWh relative to risk-adaptive residual-CVaR control. All three intervals exclude zero, and all three exact tests remain significant after Holm correction. Mean realized operating cost is 153.83–280.17 yuan higher, so the result is a reliability–cost trade-off and does not establish equal-cost or Pareto superiority.

Component interventions support residual scenarios, generator-first execution, and the supervisory package; they do not support an independent reliability effect for adaptive CVaR.

Public real measurements add bounded component evidence without closing the controller-validation gap. Eight electrical series from three dataset families support cross-domain forecasting most consistently at 15 and 60 min, and two BESS records support coherent power–SOC behavior under frozen eligibility rules. Pooled diagnostics from two drilling records, however, challenge the simulator’s state-persistence distribution; the high unknown-label fraction in Energistics precludes a like-for-like calibration claim. Within these boundaries, the results support coordinating storage reserve with forecast uncertainty and model-reachable generator capacity in the frozen simulation. Site-specific electrical replay, real-data-informed temporal recalibration, and hardware-in-the-loop testing are required before drawing field-performance conclusions.

CRediT authorship contribution statement

Shixing Zhang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing–original draft, Project administration.

Pengchong Wei: Methodology, Validation, Investigation, Data curation, Writing–review & editing.

Lei Luo: Resources, Validation, Writing–review & editing.

Pengfei Zhang: Investigation, Validation, Writing–review & editing.

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Declaration of competing interest

All authors are employees of companies affiliated with China National Petroleum Corporation (CNPC). The authors declare no other competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

The evidence-synchronized V24/V28 source and compact audit package is available under the public archival tag `v24-v28-reproducibility-20260922`. It fixes the manuscript, configurations, source code, tests, and retained compact evidence to immutable Git commits; the code and evidence submodule is pinned to commit `3499ed3`. Raw third-party datasets, prepared Parquet files, prediction exports, model weights, and bulk trajectory outputs are excluded because of size, licensing, or non-authoritative status. Dataset identities, source versions, preparation audits, cleaning rules, and retained result summaries are included. The earlier V21 package remains separately archived as `v1.0.0` at <https://doi.org/10.5281/zenodo.22739760>; it does not reproduce the V24/V28 results. A DOI for the V24/V28 package will be added after its archival deposit. The public-data sources are identified by immutable records or versioned repositories: REFIT (<https://doi.org/10.5281/zenodo.5063428>), UCI household power (<https://doi.org/10.24432/C58K54>), the Tsukuba microgrid (<https://doi.org/10.6084/m9.figshare.7403954.v1>), M5BAT Pb1 (<https://doi.org/10.18154/RWTH-2026-06637>), Utah FORGE Well 58-32 (<https://doi.org/10.15121/1495411>), the Energistics Well B example (<https://energistics.org/sites/default/files/2023-03/NA-NA-EnergisticsWell2016-B.zip>), Petrobras 3W at commit `d717f0aadbe90b6dc6e15e29f6035e9630d22a5e`, and OpenCEM at commit `5884d253a5267fb240b7a8df6fa9e4d49a905167`. The two repositories are available at <https://github.com/petrobras/3W> and <https://github.com/OpenCEM-platform/opencem-dataset>, respectively. Their original licenses govern redistribution. The package records source URLs, versions, hashes, cleaning rules, and derived one-minute artifact identities rather than relicensing the raw data.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, OpenAI Codex was used to assist with LaTeX formatting, language and structural editing, and consistency checks. The tool was not used as an author and did not determine the reported methods or results. After using the tool, the authors reviewed and edited its output and take full responsibility for the content of the publication.

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